

OCR A-Level Computer Science
Component 01: Computer Systems
1.3 Exchanging data

Centre Number:

Candidate Number:

Surname:

Forename(s):

Date:

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above.
- Answer ALL questions.
- Write your answers in the spaces provided.
- Additional paper may be used if required.
- The total mark for this paper is 40.
- The marks for each question are shown in brackets [].
- Quality of extended response will be assessed in questions marked with an asterisk (*).

INFORMATION FOR CANDIDATES

- The total time for this paper is 75 minutes.
- This document consists of 8 pages.
- Calculators may be used.

Questions

1. Compression

- (a) Define the term 'compression' as used in computer science. [2 marks]

- (b) Explain the difference between lossy and lossless compression, giving one example of a file format that uses each. [4 marks]

- (c) A user needs to compress a collection of high-resolution photographs for:

- Long-term archival storage
- Display on a website

Recommend a suitable compression method for each purpose, justifying your choices. [4 marks]

2. Encryption

- (a) Explain what is meant by encryption in computer science. [2 marks]

- (b) Describe the difference between symmetric and asymmetric encryption. [3 marks]

- (c) Explain how public and private keys are used in asymmetric encryption to send a secure message. [4 marks]

- (d) Explain why HTTPS is important for e-commerce websites. [3 marks]

3. Transaction Processing

- (a) Define the term 'transaction' in the context of database systems. [2 marks]

- (b) Explain what is meant by ACID properties in transaction processing. [4 marks]

- (c) A banking system processes transfers between accounts. Explain how the ACID properties ensure that a transfer of £100 from Account A to Account B is processed reliably. [4 marks]

4. Web Technologies

- (a) Describe the purpose of HTML, CSS, and JavaScript in web development. [3 marks]

- (b) Explain what is meant by client-side and server-side processing, giving one example of each. [4 marks]

5. *Extended Response Question

A large online retailer is redesigning their e-commerce platform. The system needs to handle customer data, process payments, and manage inventory across multiple warehouses.

Discuss the data exchange technologies and methods that would be required for this system, including:

- Appropriate compression techniques
- Security measures including encryption
- Transaction processing requirements
- Web technologies for the customer interface

For each technology or method, explain why it is appropriate and how it contributes to the overall system. [10 marks]

Grade Boundaries

Grade	A*	A	B	C	D
Marks	36-40	32-35	28-31	24-27	20-23